

Author	Year	Theme	Topic	Main Finding
Menon et al.	2025	AI-Powered IoT, Smart Applications, Security	Integration of AI and IoT (AloT) for enhanced security, efficiency, and smart systems	AloT combines IoT data collection with AI/ML/DL techniques to improve security, anomaly detection, automation, decision-making, and operational efficiency across healthcare, transportation, industry, smart homes, and smart cities.

Limitations	IoT Sensors Used	AI Technique Used	Privacy	Focused Area
Resource-constrained IoT devices, high energy consumption, scalability issues, privacy concerns, lack of interoperability, security vulnerabilities, and computational limitations for AI processing.	General IoT devices and sensors (RFID, WSN sensors, wearable sensors, smart objects, environmental sensors, healthcare monitoring sensors).	Machine Learning (ML), Deep Learning (DL), CNN, RNN, ANN, Reinforcement Learning (RL), Deep Reinforcement Learning (DRL), Federated Learning (FL), Hyperdimensional Computing.	Strong emphasis on privacy and security through AI-based intrusion detection, authentication, access control, anomaly detection, malware analysis, and blockchain integration.	IoT security, AIoT architecture, smart healthcare, transportation, industrial automation, smart cities, smart homes.

Implementation	Key Contribution
Comprehensive survey and conceptual framework; no direct deployment. Reviews existing architectures, AI techniques, and emerging technologies.	Provides a comprehensive survey of AIoT covering architectures, security mechanisms, ML/DL approaches, emerging technologies (Blockchain, 6G, FL, HD Computing), applications, challenges, and future research directions.